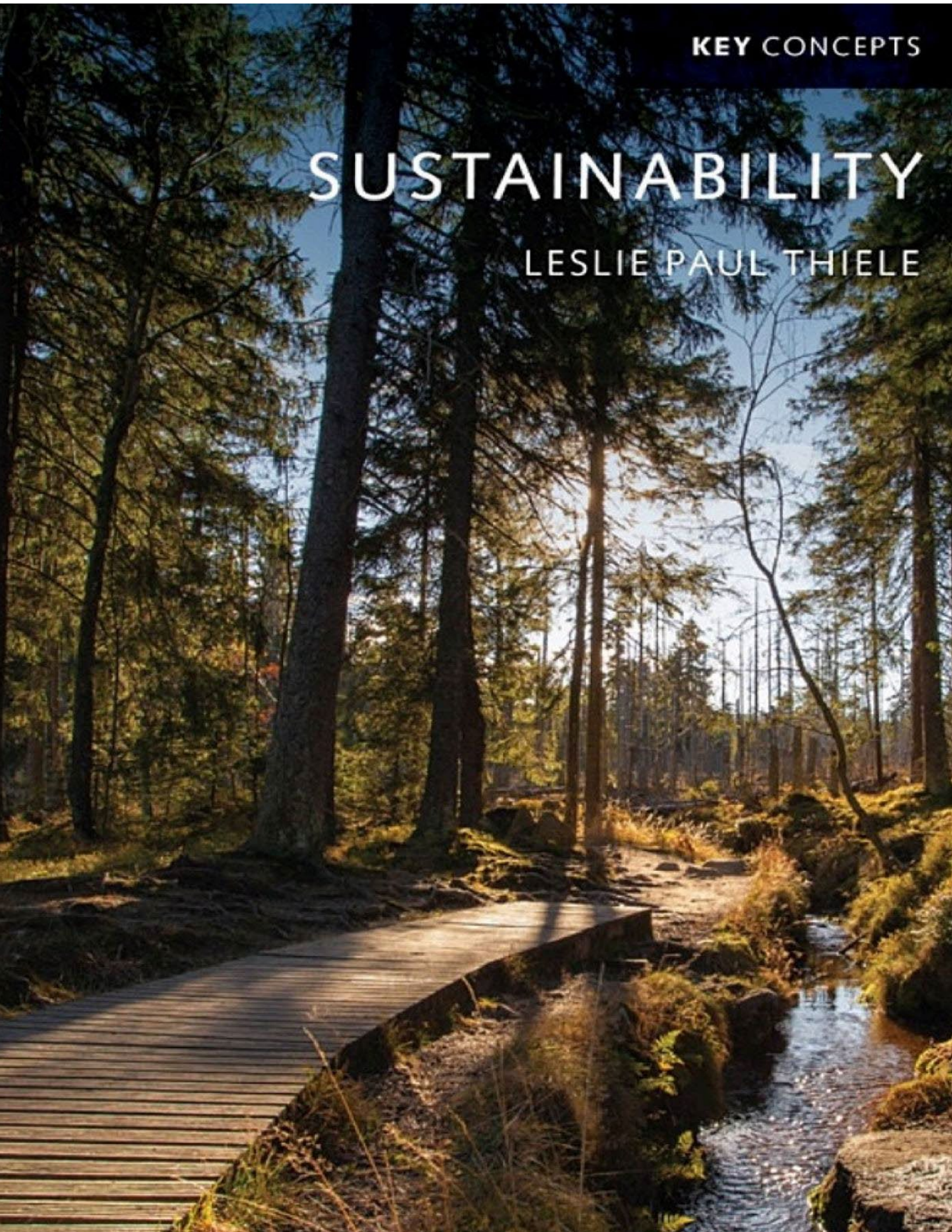
KEY CONCEPTS

SUSTAINABILITY

LESLIE PAUL THIELE



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Introduction

1 The word "sustainability" derives from the Latin *sustinere*, which literally means "to hold up." Something is sustainable if it endures, persists, or holds up over time. The motivation to live sustainably dates back to ancient times. But widespread, systematic efforts to promote sustainability are only a few decades old. In this relatively short time, the pursuit of sustainability has generated lifestyle changes for countless individuals, mission and policy changes for professional agencies, educational institutions, and civic organizations, innovations in business, design, engineering, natural resource use and agriculture, new national laws, and crucial international protocols and agreements. Sustainability has become one of a very few ideals – joining the ranks of democracy and human rights – that receive near universal endorsement. Across a diverse globe of peoples and nations, sustainability increasingly provides a common language and goal.

2 In an age of global climate change, natural resource depletion, and "failing states" that can no longer meet their people's basic needs, sustainability has been given the daunting tasks of "rescuing civilization" and "saving the planet." Learning to live and work sustainably is arguably the greatest challenge of our times.

3 But what is sustainability? Is it an ethical ideal to be striven for? Or is it the scientifically grounded effort to efficiently manage and conserve natural resources? Sustainability certainly has ethical components. It is grounded in moral claims about the responsibilities and obligations of individuals and organizations. And it finds support in age-old virtues. But sustainability is also grounded in science. It demands sound scholarship and evidence-based decision making that accounts for current conditions and trends while demonstrating how lifestyles, economies, technologies, and social practices can be effectively transformed to lower risks, prevent harms, and secure long-term benefits. In short, sustainability is an adaptive practice wedded to science in service to an ethical vision.

An adaptive practice

4 Owing to the frequency and looseness of its usage, sustainability has been called "one of the least meaningful and most overused words in the English language."[1] The best response to this sort of criticism is not to stop using the term, but to define it clearly while making its practice more measurable and impactful.

5 Sustainability is most easily defined by saying what it is not. A practice, relationship, or institution is not sustainable if it undermines the social, economic, or environmental conditions of its own viability. It is unsustainable to extract water from rivers, lakes, and aquifers at a faster rate than they can be naturally recharged by rain and snow. Doing so will produce water-starved communities. Eroding the land upon which crops grow faster than fertile soil is regenerated is not sustainable agriculture. It will produce failing farms and hunger. Running a business consistently in the red, with revenues that do not exceed expenses, is not sustainable. It will end in bankruptcy.

6 With this in mind, sustainability is often defined as meeting current needs in a way that does not undermine future welfare. But whose needs are being met and whose future welfare is at stake? At its best, sustainability extends concern beyond the welfare of those who are directly involved in a practice, relationship, or institution. It fosters concern for the welfare of all who become impacted by our actions. That includes people who are distant in time or space, as well as other species.

7 In an increasingly interdependent world, the impacts of our actions (and inactions) cross borders and generations, spanning the globe and casting long shadows into the future. The

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natural resources utilized in many of the goods we purchase have been extracted and processed in other countries. The waste and pollution generated when these goods get thrown away may end up back in far-off lands and oceans. The carbon dioxide that is emitted from factory smokestacks, energy plants, and our vehicles' tailpipes envelops the globe and will have its greatest impact on distant progeny as the planet's temperature steadily increases under a blanket of greenhouse gases while rising oceans swamp shorelines. Sustainability concerns the global, long-term impact of our practices, relationships, and institutions because we live in a connected world.

8 Sustainability is typically understood as the effort to use fewer natural resources and produce less waste. Consequently, it is often identified with recycling and energy efficiency. To be sure, it is crucial that we use natural resources wisely and avoid waste. But sustainability is both more challenging and more rewarding. It pushes us to better understand our world and ourselves, and to cultivate a sense of responsibility for the health and resilience of the ecological, social, economic, and cultural networks that support us.

9 To that end, transdisciplinarity is crucial. No single field of study can illuminate all or even most of the interconnected relationships that define our lives and world. To craft sustainable solutions, we must integrate multiple intellectual disciplines and involve multiple stakeholders, including citizens, businesses, community organizations, and governments.

10 The goal of such transdisciplinary inquiry and action, notwithstanding common misperceptions, is not simply preservation. The aim is not to maintain a "culture of permanence."[2] Sustainability requires both conservation and creativity. To be effective in our conservation efforts, we must be adaptive. Practicing sustainability entails managing well the scale and speed of change.

11 For example, global warming and global cooling have occurred many times in the history of the planet. Climate change is nothing new. However, the current level of atmospheric carbon dioxide, measured in parts per million, is growing at many times the rate that occurred the last time the earth lost its polar ice caps millions of years ago, when much of its land mass became submerged under enlarging seas.

12 The CO_2 in the atmosphere prior to the industrial age was about 280 parts per million (ppm), and it has been at that or a lower level for the last 800,000 years. James Hansen, one of the world's leading climatologists, announced in 2007 that 350 ppm of CO_2 was the maximum safe level. By 2023, 425 ppm had been reached, and the number is rising steadily. Clearly, we are in uncharted territory.

13 Our use of fossil fuels and our destruction of forests are unsustainable practices because they are causing disruptive change at a scale and speed that preempt successful adaptation by many forms of life. Along with habitat loss and pollution, global warming is already one of the top causes of extinction. Our species is altering the climate of the planet at such an accelerated rate that it will prove impossible for millions of species, quite possibly including our own, to adapt in time.

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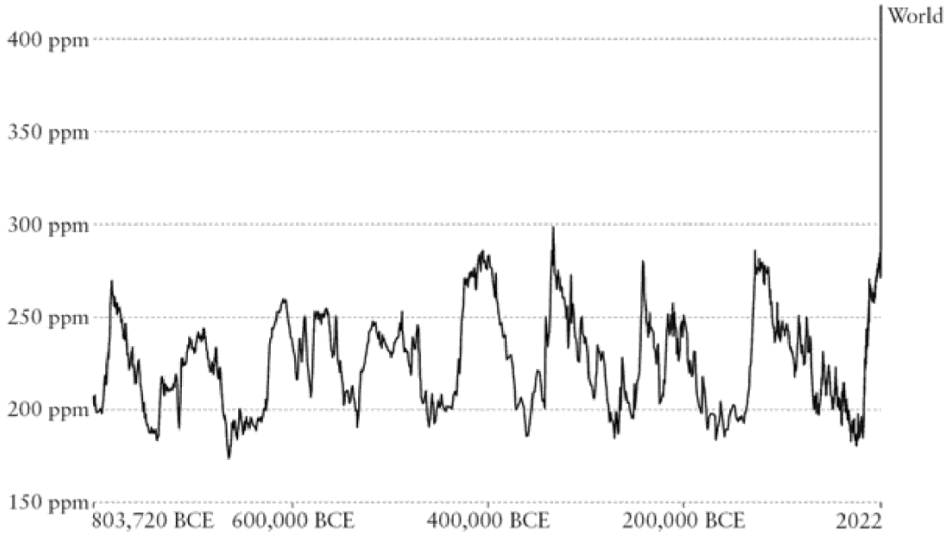

Figure 0.1: Global atmospheric CO_2 concentration

Note: Atmospheric carbon dioxide (CO_2) concentration is measured in parts per million (ppm). Long-term trends in CO_2 concentrations can be measured at high-resolution using preserved air samples from ice cores.

Source: Our World in Data and the National Oceanic and Atmospheric Administration.

14 Sustainability is the practice of satisfying current needs without sacrificing future well-being. This can only be achieved by managing the scale and speed of change in a way that conserves the core functions, values, and relationships of the communities of life that sustain us. Balancing adaptive change with conservation is achieved through the integrated pursuit of ecological resilience, economic welfare, social equity, and cultural creativity.

15 Ecological resilience entails the conservation of vibrant, biodiverse wild spaces and ecosystems while maintaining relatively pollution-free land, water, and atmospheric environments. Economic welfare entails the creation of infrastructure and opportunities for people to pursue enterprise and prosperity, with mechanisms to avoid corrosive disparities of wealth and aid those who cannot meet their basic needs. Social equity entails providing people with the means to direct their individual and collective lives, including access to education, democratic government, basic human rights, and a vigorous civil society. Equity is grounded in empowerment. Cultural creativity entails opportunities to engage, explore, enrich, and innovate all facets of human culture, including science and knowledge, ethics and politics, economy and technology, customs and diet, arts and recreation, religion and spirituality.

16 When people talk about sustainability they most often focus on environmental issues. To be truly sustainable, however, a practice, relationship, or institution must do more than protect nature and conserve natural resources. It must meet economic needs and cultivate economic opportunities. In turn, it must cultivate equitable relationships and enable empowerment. You are not running a sustainable business, no matter how "green" your practices, if you consistently fail to make a profit and cannot pay your employees. Likewise, in today's connected world, institutions and organizations that do not equitably treat and empower stakeholders are unlikely to provide enduring economic welfare and environmental health.

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17 Sustainability has traditionally been described as standing on the three pillars of society, ecology, and economy, or, alternatively, as grounded on the "triple bottom line" of people, planet, and profit. The point is that these three goods stand or fall together. You cannot have one without the other two, owing to the interdependent nature of our world. But these formulas should not be interpreted to suggest that sustainability requires a static or rigid balance of forces. By insisting on cultural creativity as a fourth component of sustainability, we underline the fact that our practices, relationships, and institutions have to initiate and respond to change if they are to endure for long. Sustainability demands imagination and innovation.

18 The humorist Will Rogers once remarked that even if you are on the right track, you will eventually get run over by a train if you just sit there. To practice sustainability well, we cannot just sit there. We have to learn and adapt. Without creativity, it is impossible to sustain what we value.

Notes

1 David Owen, *The Conundrum: How Scientific Innovation, Increased Efficiency, and Good Intentions Can Make Our Energy and Climate Problems Worse* (New York: Riverhead Books, 2011), p. 246.

2 David Selby, "The Firm and Shaky Ground of Education for Sustainable Development," *Journal of Geography in Higher Education* 30(2) (July 2006): 363.

3 Charlotte Elton, "Companies will soon have to prove that they really are taking climate action, under draft EU law," *Euronews*, January 16, 2023, accessed at: https://www.euronews.com/green/2023/01/16/companies-will-soon-have-to-prove-that-they-really-are-taking-climate-action-under-draft-e

4 Liz Ford, "Rio+20 Politicians Deliver 'New Definition of Hypocrisy' Claim NGOs," *Guardian*, June 21, 2012, accessed at: http://www.guardian.co.uk/global-development/2012/jun/21/rio20-politicians-hypocrisy-ngos?intcmp=239

5 Donald Worster, "The Shaky Ground of Sustainability," in George Sessions (ed.), *Deep Ecology for the 21st Century* (Boston, MA: Shambhala, 1995), p. 418.

6 United Nations, Press Release, April 4, 2022, accessed at: https://press.un.org/en/2022/sgsm21228.doc.htm

7 United Nations, UN News, November 7, 2022, accessed at: https://news.un.org/en/story/2022/11/1130247

8 See: http://www.earthfirstjournal.org/section.php?id=1

9 Donald Mann, president of Negative Population Growth, quoted in Garrett Hardin, *Living within Limits: Ecology, Economics, and Population Taboos* (New York: Oxford University Press, 1993), p. 206.

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